STREAMLINE Testing Data Evaluation Report: 2025-12-23 10:36:48.682298

General Pipeline Settings:

CV Partitions: 3
Partition Method: Stratified
Categorical Cutoff: 10
Statistical Significance Cutoff: 0.05
Random Seed: 42
Run From Notebook: False

Feature Importance/Selection Settings:

Use Mutual Information: False
Use MultiSURF: True
Use TURF: False
TURF Cutoff: 0.5
MultiSURF Instance Subset: 1500
Max Features to Keep: None
Filter Poor Features: True
Top Features to Display: 40
Export Feature Importance Plot: True

ML Modeling Algorithms:

Artificial Neural Network: False
Category Gradient Boosting: False
Decision Tree: True
Elastic Net: False
ExSTraCS: False
Extreme Gradient Boosting: True
Genetic Programming: False
Gradient Boosting: False
K-Nearest Neighbors: True
Light Gradient Boosting: False
Logistic Regression: True
Naive Bayes: False
Random Forest: True
Support Vector Machine: False
XCS: False
eLCS: False

Modeling Settings:

Primary Metric: balanced_accuracy
Training Subsample for KNN,ANN,SVM,and XGB: 0
Uniform Feature Importance Estimation (Models): True
Hyperparameter Sweep Number of Trials: 100
Hyperparameter Timeout: None
Export Hyperparameter Sweep Plots: False

LCS Settings (eLCS,XCS,ExSTraCS):

Do LCS Hyperparameter Sweep: False
nu: 1
Training Iterations: 2000
N (Rule Population Size): 2000
LCS Hyperparameter Sweep Timeout: 1200

Stats and Figure Settings:

Export ROC Plot: True
Export PRC Plot: True
Export Metric Boxplots: True
Export Feature Importance Boxplots: True
Metric Weighting Composite FI Plots: balanced_accuracy
Top Model Features To Display: 50

EDA and Processing Settings:

Engineering Missingness Cutoff: 1.0
Cleaning Missingness Cutoff: 1.0
Correlation Removal Threshold: 1.0
List of Exploratory Analysis Ran: ['Univariate Analysis', 'Feature Correlation']
List of Exploratory Plots Saved: []
Use Data Scaling: False
Use Data Imputation: False
Use Multivariate Imputation: False
Overwrite CV Datasets: False

Target Dataset(s):

D1 = gene_pa_merged_train

Target Data Settings:

Data Path:
/Users/921623492/Ecoli_Project/ML/Final_ML_03/STREAMLINE/data/Gene_Presence_Absence
Output Path:
/Users/921623492/Ecoli_Project/ML/Final_ML_03/output/Abdoul_thesis_ML_parallel_Gene_Presence_Absence
Experiment Name: Abdoul_thesis_ML_parallel_Gene_Presence_Absence
Class Label: label
Instance Label: sample_id
Match Label: None
Ignored Features: None
Specified Categorical Features: None
Specified Quantitative Features: ['gene_presence_absence.csv']

Univariate Analysis of Each Dataset (Top 10 Features for Each)

D1 = gene_pa_merged_train

Feature	p-value	Test-statistic	Test-name
MAOGMHBA_04934_pemI	1.6267085125697681e-38	168.43352306909156	Chi Square Test
MAOGMHBA_04935_pemK	1.5111292878002085e-37	164.0021059893611	Chi Square Test
MAOGMHBA_04933_group_2527	4.538068102003527e-37	161.81607669222876	Chi Square Test
GHHBFBJJ_04052_sopB	7.403260048995429e-37	160.84319644136724	Chi Square Test
GHHBFBJJ_04053_group_39082	1.347141208689208e-36	159.653233576703	Chi Square Test
FAHFDEJI_01302_tabA_2	5.846419865872208e-36	156.7357607578113	Chi Square Test
ECKICHIN_02367_bdcA	8.470241167530789e-36	155.9989553617146	Chi Square Test
ECKICHIN_02368_yjgH	8.470241167530789e-36	155.9989553617146	Chi Square Test
FAHFDEJI_01300_arcC1_3	2.594381498201355e-35	153.7743898824209	Chi Square Test
ECKICHIN_02363_group_26113	5.429916492947158e-35	152.30670746606552	Chi Square Test

Dataset and Model Prediction Summary: D1 = gene_pa_merged_train

Data Processing/Counts Summary

	Instances	Total Features	Categorical Features	Quantitative Features	Missing Values	Missing Percent	Class 0	Class 1
Original	470	6092	6092	0	0	0.0	245	225
C1	470	6092	6092	0	0	0.0	245	225
E1	470	6092	6092	0	0	0.0	245	225
C2	470	3519	3519	0	0	0.0	245	225
C3	470	3519	3519	0	0	0.0	245	225
E2	470	3519	3519	0	0	0.0	245	225
C4	470	3082	3082	0	0	0.0	245	225
Processed	470	3082	3082	0	0	0.0	245	225

Cleaning (C) and Engineering (E) Elements

* C1 - Remove instances with no outcome and features to ignore
* E1 - Add missingness features
* C2 - Remove features with invariance or high missingness
* C3 - Remove instances with high missingness
* E2 - Add one-hot-encoding of categorical features
* C4 - Remove highly correlated features

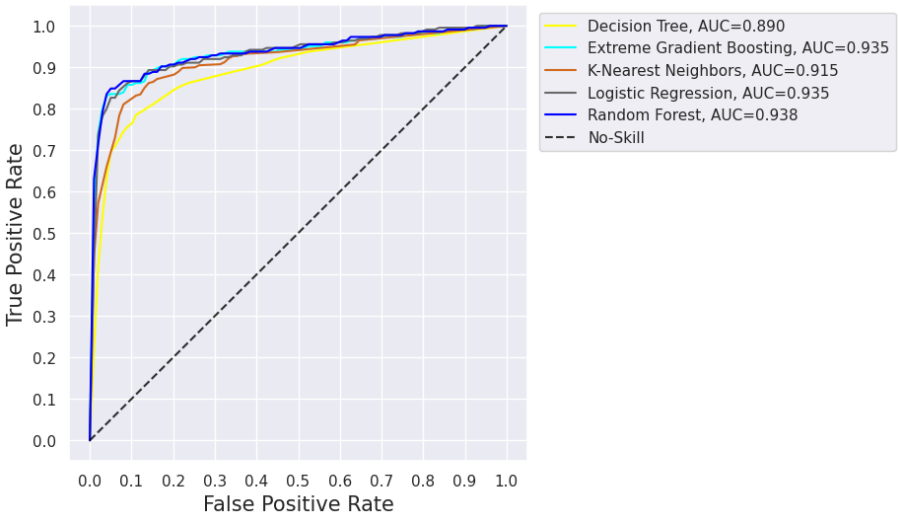
Class Balance (Processed)

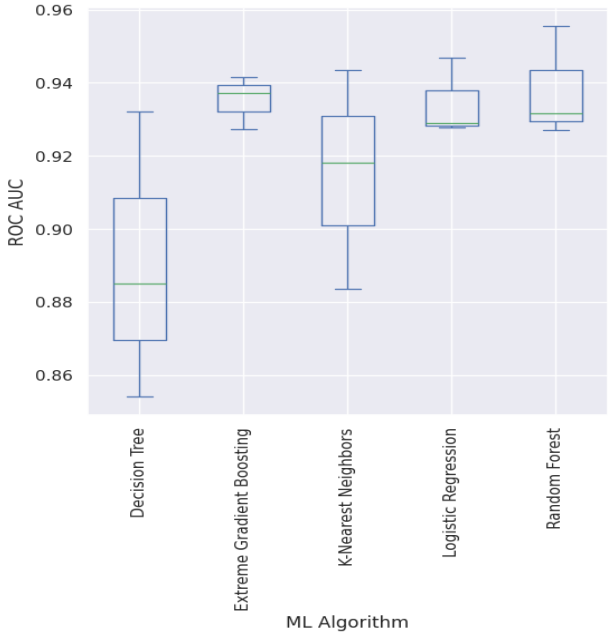
Feature Correlations (Pearson)

No Feature Correlation Plot

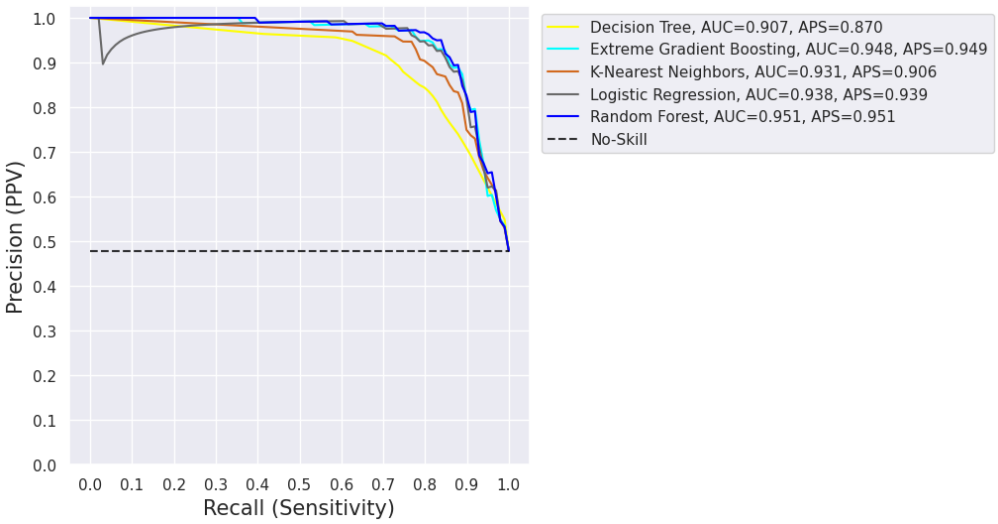
Top ML Algorithm Results (Averaged Over CV Runs):
Best (ROC_AUC): Random Forest = 0.938
Best (Balanced Acc.): Random Forest = 0.898
Best (F1 Score): Random Forest = 0.890
Best (PRC AUC): Random Forest = 0.951
Best (PRC APS): Random Forest = 0.951

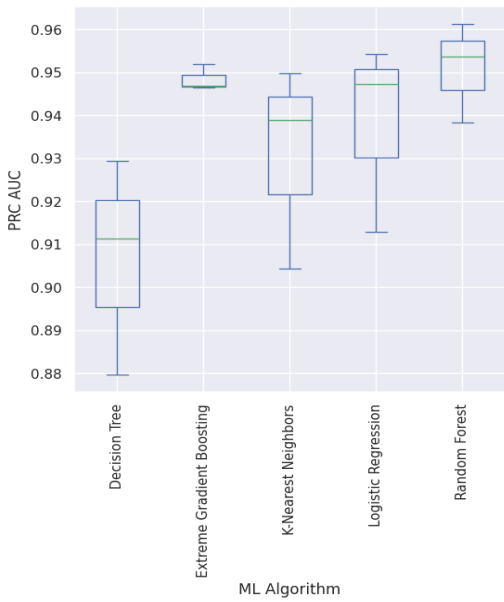
ROC





PRC





Generated with STREAMLINE (0.3.4): (https://github.com/UrbsLab/STREAMLINE)

Average Model Prediction Statistics (Rounded to 3 Decimal Points)

D1 = gene_pa_merged_train

ML Algorithm	Balanced Accuracy	Accuracy	F1 Score	Sensitivity (Recall)	Specificity	Precision (PPV)	TP	TN	FP	FN	NPV	LR+	LR-	ROC AUC	PRC AUC	PRC APS
Decision Tree	0.838	0.84	0.823	0.778	0.898	0.875	58.333	73.333	8.333	16.667	0.816	7.738	0.248	0.89	0.907	0.87
Extreme Gradient Boosting	0.894	0.896	0.886	0.849	0.939	0.928	63.667	76.667	5.0	11.333	0.871	16.427	0.162	0.935	0.948	0.949
K-Nearest Neighbors	0.863	0.866	0.851	0.804	0.922	0.906	60.333	75.333	6.333	14.667	0.838	10.696	0.212	0.915	0.931	0.906
Logistic Regression	0.882	0.883	0.875	0.858	0.906	0.894	64.333	74.0	7.667	10.667	0.874	9.584	0.157	0.935	0.938	0.939
Random Forest	0.898	0.9	0.89	0.844	0.951	0.941	63.333	77.667	4.0	11.667	0.869	17.983	0.164	0.938	0.951	0.951

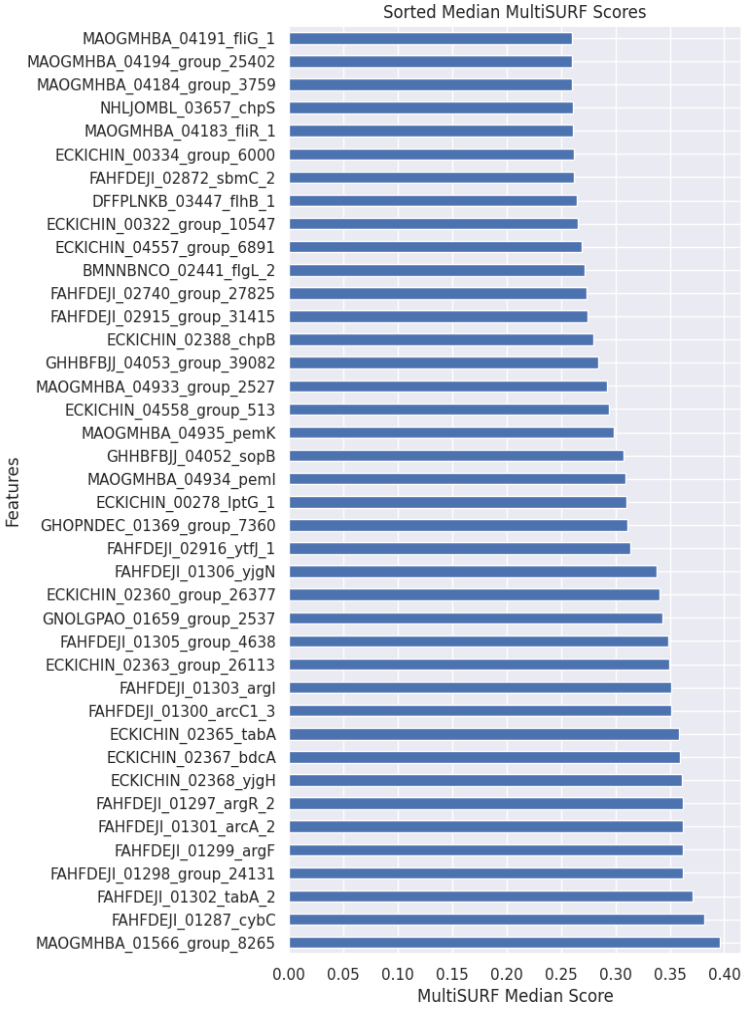
Median Model Prediction Statistics (Rounded to 3 Decimal Points)

D1 = gene_pa_merged_train

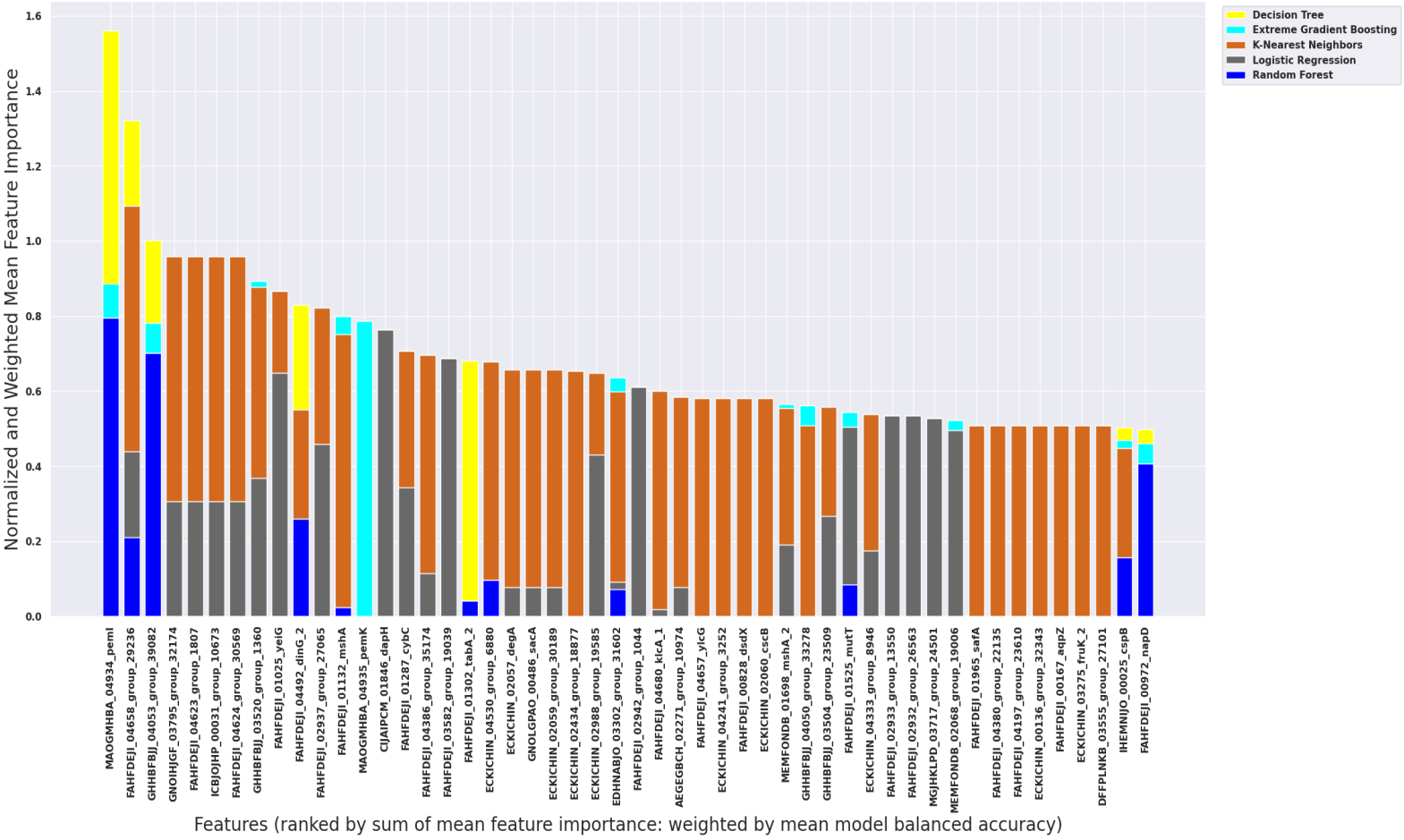
ML Algorithm	Balanced Accuracy	Accuracy	F1 Score	Sensitivity (Recall)	Specificity	Precision (PPV)	TP	TN	FP	FN	NPV	LR+	LR-	ROC AUC	PRC AUC	PRC APS
Decision Tree	0.851	0.854	0.837	0.787	0.89	0.873	59.0	73.0	9.0	16.0	0.824	7.44	0.233	0.885	0.911	0.859
Extreme Gradient Boosting	0.908	0.91	0.901	0.853	0.951	0.942	64.0	78.0	4.0	11.0	0.876	17.767	0.152	0.937	0.947	0.947
K-Nearest Neighbors	0.863	0.866	0.849	0.787	0.926	0.906	59.0	75.0	6.0	16.0	0.828	10.44	0.227	0.918	0.939	0.925
Logistic Regression	0.884	0.885	0.877	0.853	0.915	0.901	64.0	75.0	7.0	11.0	0.872	9.996	0.16	0.929	0.947	0.948
Random Forest	0.901	0.904	0.894	0.84	0.951	0.941	63.0	78.0	4.0	12.0	0.867	17.493	0.166	0.932	0.954	0.954

Feature Importance Summary: D1 = gene_pa_merged_train

Mutual Information Plot Not Available



Composite Feature Importance Plot (Normalized and Performance Weighted)



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Compare ML Performance Across Datasets

Using Best Performing Algorithms (Kruskall Wallis Compare Datasets)

Pipeline Runtime Summary

gene_pa_merged_train

Pipeline Component	Phase	Time (sec)
Exploratory Analysis	1	140.98094511032104
Scale and Impute	2	-168.37193131446838
MultiSURF (Feature Importance)	3	259.734548330307
Feature Selection	4	0.9807939529418944
Decision Tree(Modeling)	5	231.0222058296204
Extreme Gradient Boosting(Modeling)	5	855.6664798259735
K-Nearest Neighbors(Modeling)	5	37385.78645372391
Logistic Regression(Modeling)	5	547.7318315505981
Random Forest(Modeling)	5	2594.406277656555
Stats Summary	6	11.218124866485596